

Cluster Robust Variance Estimation

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Table of contents

1 Introduction 1

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In constructing the confidence intervals of the OLS estimator $\hat{\beta}_{\text{ols}}$, it is the asymptotic distribution of

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N X_i u_i$$

that determines the estimator's asymptotic distribution. Assuming that $E[X_i u_i] = 0$, the Lindernberg-Levy CLT apply:

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N X_i u_i \xrightarrow{d} \mathcal{N} \left(0, \frac{1}{N} \sum_i \text{Var} (X_i u_i) \right)$$

But this hold true only if the individual observations are all independent, in fact summing 2 random variables X_1 and X_2 we get another random variable (call it Z) with expected value equal to $E[X_1] + E[X_2]$ and variance equal to $\text{Var} (X_1) + \text{Var} (X_2) + 2 \cdot \text{cov} (X_1, X_2)$. Summing N random variables $(X_1, \dots, X_n)$, assume $E[X_i] = 0$, we get a new random variable $Z_N = \sum_{i=1}^N X_i$:

$$Z_N \sim \left(0, \sum_{i=1}^N \text{Var} (X_i) + 2 \sum_{i \neq j}^N \sum_j^N \text{cov} (X_i, X_j) \right)$$

if all the X_i s are **independent** all the joint moments (and so also the convariancers) are equal to zero and, dividing by $\sqrt{N}$:

$$\frac{1}{\sqrt{N}} (Z_N) \sim \left(0, \frac{1}{N} \sum_{i=1}^N \text{Var} (X_i) \right)$$

So the variance of Z_N scaled by $\frac{1}{\sqrt{N}}$ is the average variance of all the X_i s. But here comes the problem, if our data are clustered we do not have that all the observation are independent, but rather we have that the observation of all the individuals inside a cluster are correlated, so we would have that

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N X_i u_i \xrightarrow{d} \mathcal{N} \left(0, \frac{1}{N} \sum_{i=1}^N \text{Var} (X_i u_i) + \frac{2}{N} \sum_{i \neq j}^N \sum_j^N \text{cov} (X_i u_i, X_j u_j) \right)$$

The White variance estimator $\xrightarrow{d} \frac{1}{N} \sum_{i=1}^N \text{Var} (X_i u_i)$, so it would not consistently estimate the asymptotic variance.